Bit-Band Operations – Made Simple

August 20, 2025

Contents

1 Introduction **1**
1.1 Bit-banding 1

2 Bit-band address calculation **2**
2.1 Example 2

3 Advantage of bit-band operations **3**

4 Exercises **3**

1 Introduction

The Cortex-M processors have 32 bits for addressing memory, leading to a memory space of 4GB. This 4GB memory space is architecturally divided into a number of regions. Please refer to the memory map in the reference manual. Cortex-M4 processors support both little-endian and big-endian memory systems. STM32F401/411 is little endian. The memory is byte-addressed. The memory is viewed as a serial ordering of words (4 bytes), with the address range 0x00000000 – 0xFFFFFFFF.

1.1 Bit-banding

In general, the memory is byte addressed. There are two predefined regions in the memory map that allows access to a single data bit. These regions are called *bit-band regions*. One of them is located in the first 1MB of the SRAM region, and the other is located in the first 1MB of the peripheral region. These memory regions can also be accessed like normal regions. Single data bit access is done via a separate region called the *bit-band alias*. The addresses in the *bit-band alias* region are remapped to single data bits in the *bit-band regions*.
There are two regions of memory for bit-band operations as shown in the figure 1.

- 0x20000000 – 0x200FFFFF (SRAM, 1MB)
- 0x40000000 – 0x400FFFFF (Peripherals, 1MB)

Access to the *bit-band alias* will cause an access to the *bit-band regions*. A bit band operation performs an access to the *bit-band region*.
The figure 2 below shows the mapping of bit-band alias addresses to the single data bits in the bit-band region.

The bit-band region of the peripheral memory region can be accessed via the bit-band aliased addresses as shown in the figure 3.

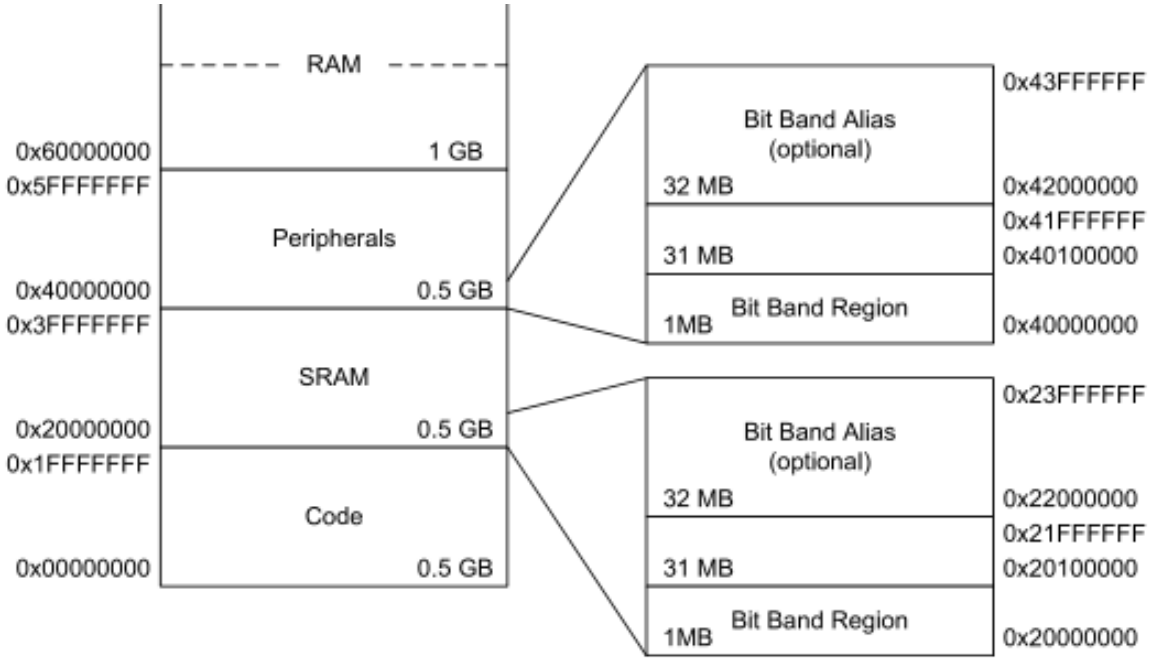


Figure 1: Pre-defined memory map of the Cortex-M4 Processor, showing the bit-band areas)

2 Bit-band address calculation

The bit-band address is calculated with respect to the first address, known as the base address. Consider the bit-band region in the peripheral address space. The base address of the bit-band peripheral area is 0x40000000, which is mapped to the bit-band alias base address of 0x42000000.
Also, note that the address of the nth bit position is obtained by:
Bit-band alias word address + n*4.
This can easily be understood from the table above. The bit-band alias word address is found by adding 32 * offset value to the bit-band alias base, as each word in the bit-band region consumes 32 addresses, which are multiples of 4.

2.1 Example

Calculate the bit-band address of the 13th *bit in the PORTC's ODR register (PC13).*
Please open the reference manual.
From page 38, we find that the base address of PORTC is 0x4002 0800.
Also, from page 165, we find that the ODR is at an offset of 0x14.
Hence, the address of PORTC ODR register is = PORTC base address + ODR Offset
That is, PORTC ODR address = 0x4002 0800 + 0x14 = 0x4002 0814.
Next, we need to compute the offset of PORTC ODR with respect to the peripheral base address 0x40000000.
Offset from the peripheral base = 0x4002 0814 - 0x4000 0000 = 0x20814
Next, to find the bit-band alias word address, we need to multiply this offset by 32 and add it to the bit-band alias base address.
That is, bit-band alias word address = Bit-band alias base + 32 * offset.
We need to find 32 * 0x20814
We know that $32 = 2^5$. To multiply by 32, just left shift 0x20814, by 5 bits.
$00100000100000010100 << 5 = 00100000100000001010000000 = 0x0410280$
Hence, bit-band alias word address = 0x4200 0000 + 0x0410280 = 0x4241 0280

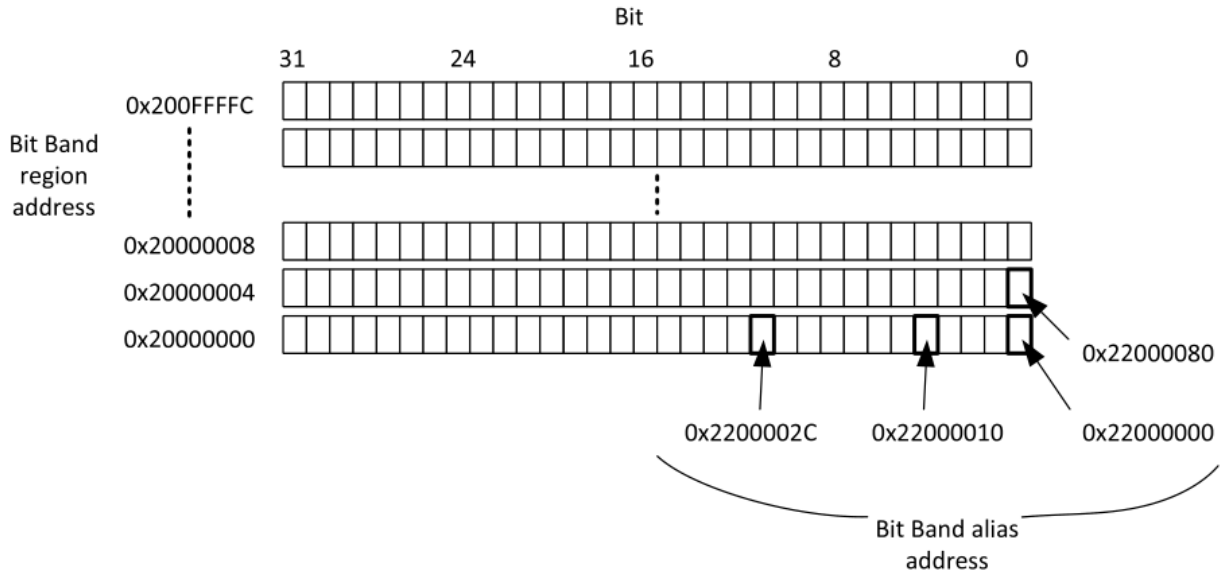


Figure 2: Bit accesses to bit-band region via the bit-band alias(SRAM region)

$\therefore$ the bit band address of PORTC ORD 13th bit (PC13) = bit-band alias word address + (4*13)
That is 0x4241 0280 + 52(decimal) = 0x4241 0280 + 0x34 = 0x4241 02B4.
This address can ne used to read/write PC13.

3 Advantage of bit-band operations

Support for bit-band operations simplify application code. For example, if a branch should be carried out based on one singe bit in a status register in a peripheral, instead of :

- Reading the whole register
- masking the unwanted bits
- comparing and branching

We can simplify the operation to :

- Reading the status bit via the bit band alias
- Comparing and branching

Provides faster bit operations with fewer instructions. Allows for the sharing of port pins by more than one process. Bit-band operations are atomic. A software-based READ-MODIFY-WRITE operation can cause data conflicts, whereas a bit-band cannot be interrupted. Also used in handling Boolean data in the SRAM region.

4 Exercises

1. Compute the bit-band address for bit 0 at address 0x20000004.
2. Compute the bit-band address for PB10.

Remapping of Bit-Band Addresses in the Peripherals Region	
Bit-Band Region	**Aliased Equivalent**
0x40000000 bit [0]	0x42000000 bit [0]
0x40000000 bit [1]	0x42000004 bit [0]
0x40000000 bit [2]	0x42000008 bit [0]
…	…
0x40000000 bit [31]	0x4200007C bit [0]
0x40000004 bit [0]	0x42000080 bit [0]
…	…
0x40000004 bit [31]	0x420000FC bit [0]
…	…
0x400FFFFC bit [31]	0x43FFFFFC bit [0]

Figure 3: Bit accesses to bit-band region via the bit-band alias(SRAM region)

References

[1] `https://www.st.com/resource/en/reference_manual/rm0368-stm32f401xbc-and-stm32f401xde-advanced-armbas pdf`. [Accessed 20-08-2025].

[2] Joseph Yiu. *The Definitive Guide to ARM Cortex-M3 and Cortex-M4 Processors, Third Edition*. Newnes, USA, 3rd edition, 2013. `isbn:0124080820`.